

JOHNSTON, NC, USA

WMO: 722131

Lat: 35.541N

Lon: 78.390W

Elev: 50

StdP: 100.72

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-4.1	-20.9	0.6	1.8	-16.1	0.9	3.2	8.1	12.0	7.2	13.4	0.6	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.9	24.5	33.0	24.2	32.4	23.9	27.0	30.1	26.2	29.8	25.6	29.6	2.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	27.9	25.2	20.4	27.5	24.7	19.9	27.3	84.9	30.9	81.6	29.4	78.6	30.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.0	5.6	4.8	-10.6	36.8	2.5	1.5	-12.4	37.9	-13.9	38.8	-15.3	39.6	-17.1	40.6		
			-11.0	27.8	2.4	0.9	-12.7	28.4	-14.1	28.9	-15.5	29.5	-17.2	30.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	5.7	7.4	11.0	16.4	20.9	25.0	26.8	25.9	22.9	16.7	10.6	7.6
	DBStd	8.55	5.51	5.22	5.24	4.31	3.69	2.57	1.98	2.20	2.94	4.57	4.60	5.12
	HDD10.0	478	155	104	53	4	0	0	0	0	0	4	48	110
	HDD18.3	1696	391	308	232	87	20	0	0	0	4	85	236	333
	CDD10.0	2838	23	32	85	195	337	451	520	492	388	213	65	37
	CDD18.3	1013	1	2	6	28	99	201	262	234	142	35	3	2
	CDH23.3	9466	2	16	81	327	896	1937	2623	2171	1091	292	24	7
	CDH26.7	3831	0	1	8	81	292	844	1226	943	371	64	2	0
Wind	WSAvg	1.4	1.5	1.8	1.9	2.0	1.5	1.3	1.2	0.9	1.3	1.3	1.4	1.3
Precipitation	PrecAvg	1235	101	95	112	84	106	111	150	122	115	82	77	78
	PrecMax	1738	202	166	197	222	233	378	375	270	460	394	173	186
	PrecMin	856	22	13	30	13	33	26	34	25	0	0	8	10
	PrecStd	202	42	45	45	44	53	69	78	58	81	69	42	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	25.9	27.6	31.4	33.0	36.1	36.4	36.2	32.8	31.2	26.2	24.2
		MCWB	18.2	17.8	16.3	17.7	21.7	24.3	24.4	24.7	23.7	22.8	18.4	19.6
	2%	DB	21.0	22.5	25.9	28.7	31.3	33.9	35.0	33.9	32.1	27.9	23.0	22.1
		MCWB	17.3	15.5	16.0	17.7	21.4	24.0	24.9	24.6	23.6	21.1	17.2	18.5
	5%	DB	17.9	20.0	23.0	27.3	29.9	32.5	33.1	32.6	30.9	27.0	21.3	19.7
		MCWB	14.3	14.7	15.1	17.7	21.1	23.4	24.9	24.3	22.8	20.6	17.0	16.9
	10%	DB	15.9	17.4	21.1	25.1	27.8	31.3	32.4	31.9	28.9	24.8	19.1	17.2
		MCWB	12.4	12.8	14.5	17.5	20.4	23.1	24.5	24.2	22.3	20.2	15.3	14.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.0	19.4	19.4	22.9	24.4	26.4	28.2	27.7	27.2	25.2	22.2	21.8
		MCDB	21.3	22.8	22.9	25.8	28.1	30.8	31.1	30.1	28.0	26.7	23.9	23.0
	2%	WB	17.8	17.7	18.4	21.2	23.4	25.6	27.2	26.8	25.7	23.9	19.7	19.5
		MCDB	19.5	20.5	21.9	24.6	27.4	30.7	30.7	29.9	28.2	26.4	21.6	21.3
	5%	WB	15.6	16.1	17.4	20.1	22.7	25.0	26.4	26.0	24.9	22.7	17.9	17.3
		MCDB	17.7	18.9	20.0	23.7	26.7	30.1	30.1	29.7	28.2	25.2	19.6	18.1
	10%	WB	12.7	13.5	15.9	18.9	22.1	24.3	25.7	25.4	24.0	21.2	16.4	14.7
		MCDB	14.6	16.5	20.1	23.2	26.2	29.1	29.9	29.4	27.4	23.4	19.0	16.8
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.6	13.4	13.8	12.1	11.5	11.0	10.6	10.8	12.5	13.3	11.9
		MCDBR	15.5	15.9	16.3	16.0	13.9	12.9	12.1	12.1	12.8	13.8	15.3	14.0
		MCWBR	11.4	10.1	8.0	6.9	5.4	4.8	4.4	4.4	5.3	7.4	9.5	10.3
	5% WB	MCDBR	13.8	13.7	12.7	11.5	11.0	11.2	10.9	10.2	9.7	10.6	11.6	12.4
		MCWBR	11.9	10.6	8.3	6.4	4.8	4.6	4.7	4.5	4.7	6.2	8.9	10.8
Clear-Sky Solar Irradiance	taub	0.319	0.336	0.367	0.411	0.462	0.508	0.547	0.533	0.427	0.379	0.337	0.320	
	taud	2.479	2.425	2.374	2.250	2.146	2.063	1.969	1.999	2.293	2.403	2.479	2.513	
	Ebn at Noon	882	904	902	874	830	790	755	757	829	844	853	858	
	Edh at Noon	84	100	114	136	153	166	182	173	123	101	84	77	
All-Sky Solar Radiation	RadAvg	2.60	3.28	4.32	5.46	6.03	6.42	6.11	5.49	4.60	3.72	2.86	2.27	
	RadStd	0.17	0.38	0.30	0.37	0.45	0.42	0.36	0.29	0.36	0.48	0.23	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+113

Nomenclature: See separate page